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| **Fiscal Year:** | FY 2025 | **Task Last Updated:** | FY 07/11/2025 |
| **PI Name:** | Flynn-Evans, Erin E. Ph.D. | | |
| **Project Title:** | Sleep and Wake Countermeasures for Artemis | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H) | | |
| **Human Research Program Risks:** | (1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders<br>(2) **Sleep**:Risk of Performance Decrements and Adverse Health Outcomes Resulting from Sleep Loss, Circadian Desynchronization, and Work Overload | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | erin.e.flynn-evans@nasa.gov | **Fax:** | FY |
| **PI Organization Type:** | NASA CENTER | **Phone:** | 650-279-3459 |
| **Organization Name:** | NASA Ames Research Center | | |
| **PI Address 1:** | Fatigue Countermeasures Group | | |
| **PI Address 2:** | Human Systems Integration Division, Code 262-4 | | |
| **PI Web Page:** | | | |
| **City:** | Moffett Field | **State:** | CA |
| **Zip Code:** | 94035 | **Congressional District:** | 18 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | Directed Research |
| **Start Date:** | 06/30/2024 | **End Date:** | 06/30/2026 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 8 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 4 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Whitmire, Alexandra | **Contact Phone:** | |
| **Contact Email:** | alexandra.m.whitmire@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | End date changed to 06/30/2026 per L. Juliette/JSC (Ed., 1/29/26).<br>End date changed to 12/31/2025 per L. Juliette/JSC (Ed., 6/20/25). | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | Directed Research | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Astronaut crews have historically averaged less than the recommended 7-8 hours of sleep per night and short sleep durations have been linked to poorer performance while inflight. In recent years, International Space Station (ISS) crew have had more stable schedules, with consistent bed and wake times, and private crew quarters that afford an optimized sleep environment. They also have access to countermeasures (CM) like circadian lighting, caffeine, and sleep promoting medication, which are used as needed, particularly when infrequent uncontrollable operational events require crewmembers to shift their sleep. It is becoming apparent that Artemis crews will be unlikely to have stable schedules, private sleep quarters, or optimized lighting in the vehicles due to mission constraints. In addition, due to the confined environment, shift work schedules may result in awake crewmembers disrupting sleeping crewmembers. These challenges make it important to evaluate the efficacy of CMs designed to acutely promote sleep or waking at different circadian phases.<br>Aims: 1. Test if pink noise can improve sleep outcomes and subsequent alertness. 2. Test if pink noise-induced sleep improves alertness and performance during the night either alone or in combination with blue-enriched lighting and a nap. 3. Test if blue-enriched lighting and a nap can improve alertness and performance during the night. |
| **Rationale for HRP Directed Research:** | The results of the Sleep and Wake Countermeasures (SWCM) Artemis task are needed to inform fatigue countermeasures (CMs) for Artemis missions. This research is directed because it contains highly constrained research. |
| **Research Impact/Earth Benefits:** | The benefits to life on Earth will depend on the final outcome of the study. If successful, our sleep-promoting countermeasure has the potential to help individuals, such as shift workers, sleep better during the day. It also has the potential to provide airline pilots and passengers with a countermeasure to assist with sleep while on an airplane. If effective, the wake-promoting countermeasures that we are testing could be used by shift workers and those who must work or be awake overnight. |
| **Task Progress:** | This study is a randomized counterbalanced laboratory study examining sleep and wake-promoting countermeasures for future Lunar and Mars missions. We aim to test whether pink noise can improve sleep outcomes during a daytime sleep episode and thereby subsequent alertness. We further aim to test whether pink noise-induced sleep improves alertness and performance during the night alone or in combination with blue-enriched lighting and a nap. Finally, we will test whether blue-enriched lighting and a nap can improve alertness and performance during the night.<br>The proposed study involves two laboratory visits separated by at least one week. Participants keep a regular schedule with 8.5 hours in bed each night for 2-3 weeks (at least one week before each visit). Participants are randomized to complete one intervention visit and one placebo visit. There are two interventions that participants will experience during the study, sleep-promoting (pink noise) and wake-promoting (blue-enriched lighting plus a nap). These will be delivered separately and together depending on randomization.<br><br>To date, 26 participants have completed both study visits, and four have completed one study visit. Analysis will commence upon completion of the study. |
| **Bibliography Type:** | Description: (Last Updated: 08/01/2025) |
| **Abstracts for Journals and Proceedings** | Hilditch CJ, Jansen RA, Pradhan S, Bathurst NG, Tripp MK, Glaros ZL, Arsintescu L, Gregory KB, Kirkley C, Flynn-Evans EE. "Neurobehavioral performance following awakening from a 1-hour nighttime nap in blue-enriched light." 2025 Inter-agency Federal Fatigue Management & Research Working Group.<br>Abstracts. 2025 Inter-agency Federal Fatigue Management & Research Working Group. , Jul-2025 |
| **Abstracts for Journals and Proceedings** | Hilditch CJ, Jansen RA, Pradhan S, Bathurst NG, Glaros ZL, Arsintescu L, Kirkley C, Flynn-Evans EE. "Neurobehavioral outcomes immediately following a daytime sleep opportunity with pink noise." 2025 SLEEP, Seattle, Washington, June 7-11, 2025.<br>Abstracts. 2025 SLEEP, Seattle, Washington, June 7-11, 2025. , Jun-2025 |
| **Abstracts for Journals and Proceedings** | Hilditch CJ, Jansen RA, Flynn-Evans EE. "Self-rated performance immediately following a 1-hour nighttime nap." 2025 SLEEP, Seattle, Washington, June 7-11, 2025.<br>2025 SLEEP, Seattle, Washington, June 7-11, 2025. , Jun-2025 |
| **Abstracts for Journals and Proceedings** | Jansen RA, Hilditch CJ, Flynn-Evans EE. "Perceived alertness moderates the relationship between actual and perceived performance." 2025 SLEEP, Seattle, Washington, June 7-11, 2025.<br>Abstracts. 2025 SLEEP, Seattle, Washington, June 7-11, 2025. , Jun-2025 |
| **Abstracts for Journals and Proceedings** | Hilditch CJ, Jansen RA, Pradhan S, Bathurst NG, Tripp MK, Glaros ZL, Arsintescu L, Gregory KB, Kirkley C, Flynn-Evans EE. "Neurobehavioral performance following awakening from a 1-hour nighttime nap in blue-enriched light." 94th Annual Scientific Meeting of the Aerospace Medical Association, Atlanta, Georgia, June 1-6, 2025.<br>Abstracts. 94th Annual Scientific Meeting of the Aerospace Medical Association, Atlanta, Georgia, June 1-6, 2025. , Jun-2025 |
| **Abstracts for Journals and Proceedings** | Flynn-Evans EE, Glaros ZL, Hilditch CJ, Bathurst N, Tripp M, Arsintescu L, Gregory KB, Kirkley C, Pradhan S, Jansen RA. "Countermeasures to enhance sleep and alertness for Artemis missions." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025.<br>Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025 |